MRISB 4-STEP LESSON DESIGN | UbD × ETEC

OUTCOME → PROOF → PREPARE → RESPOND → REFLECT *Plan the learning first; let the checklist follow. Click a field to type · click a box to tick it.*

Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 7 (4-8 Oct 2026) - Period 1	Ch 8 - 8.1 Programming concepts

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 7 / Lesson 1 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Ch 7 review test: trace tables strong; ~35% weak at stating an algorithm's purpose. Today: start from a 3-line trace to link algorithms to real code.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students declare and use variables and constants, choose meaningful identifiers, and run a first Python program from Cambridge pseudocode. Syllabus 0478 8.1.1; AO1, AO2.

Success criteria — *what will successful learning look / sound like?*

I can explain variable vs constant; I can write DECLARE and CONSTANT statements in pseudocode; I can translate them into Python and run the program without errors.

Key knowledge / skill / understanding:

Variable = named memory location that can change; a constant stays fixed while the program runs.

Likely misconception / access or challenge need:

'=' in Python is assignment, not equality; a constant is just a variable written in capitals. EAL: 'declare', 'assign', 'identifier'.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Think-pair-share **Time:** 8 min

What students will do:

Trace a 3-line pseudocode snippet (Score <- 5; Score <- Score + 2; OUTPUT Score). Predict output alone, compare with a partner, then share why Score changed but HourlyRate cannot.

Evidence tool: Mini-whiteboards

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Slide with snippet; mini-whiteboards; Ch 8 key-term card.

Support (low performers / SEN):

Key-term card with picture of 'labelled boxes' for variables.

Stretch (fast learners / gifted):

Explain why constants make code easier to update.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Modelling / worked exam **Time:** 22 min

What students will do:

Follow a worked example: write DECLARE/CONSTANT pseudocode for a 'school canteen order' (Price, Quantity, VAT_RATE), then code it in Python (IDLE/Thonny), run and fix errors.

Evidence tool: Practical / performance task

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Coursebook (CUP 9781108915144) 8.1; Python 3 + Thonny; starter file canteen.py.

Support (low performers / SEN):

Starter file with comments showing where each line goes.

Stretch (fast learners / gifted):

Add a second constant and output a formatted total with 2 decimal places.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Gamification **Time:** 7 min

What students will do:

Quizizz 'Variable or constant?' and 'Good or bad identifier?' (e.g. x, TotalCost, 2ndName, VAT_RATE).

Evidence tool: Digital quiz (Forms / Kahoot / Quizizz)

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☑ Individual ☐ Pair ☐ Group

Resources:

Quizizz with 10 items.

Support (low performers / SEN):

Read-aloud mode; picture prompts.

Stretch (fast learners / gifted):

Bonus: rewrite 3 bad identifiers correctly.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Hook: a phone contact list - name stays, number may change. Model on the board: DECLARE Price : REAL, CONSTANT VAT_RATE <- 0.15, then the Python equivalent. Think aloud on choosing identifiers. Show Thonny: run, read the error message, fix. Key words: variable, constant, identifier, assignment.

What is new or innovative in this lesson? *(ETEC looks for this)*

Every program starts as Cambridge pseudocode, then becomes Python, so students practise the exam notation and real coding side by side.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Spaces in identifiers; using = in pseudocode instead of <-; forgetting quotes for strings; NameError from typos; students copying without running. Likely: 'a constant is a variable in capitals' - ask what stops it changing.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit ticket (Forms, 3 Qs): define constant; correct a faulty DECLARE line; name one benefit of meaningful identifiers. Time for learners' questions.

Homework / platform *(e.g. Madrasati)*

Teams/Madrasati: coursebook 8.1 questions 1-3; write pseudocode + Python for a 'bus fare' program with one constant.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Canteen context made sense to all; EAL learners used the key-term card well. 2 of 3 classes ran the program by minute 30. Exit tickets: 80% defined constant correctly.

What will I change next time?

Install-check Thonny before class (3 PCs in 10BR2 missing it). Show the <- vs = difference twice; it caused most errors.

Students needing follow-up:

10BR3: 4 students still mixing up assignment and equality; recent arrivals need the bilingual key-term card again.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

Literacy	Numeracy	Digital	Digital research	Real-life ①	Real-life ②	Inquiry	Thinking
read + write · 2-1-1-6	numbers / data · 2-1-1-6	students use it · 2-1-1-4	trusted sources · 2-1-1-4	application · 2-1-1-5	application · 2-1-1-5	experiment / research · 2-1-1-7	problem-solving · 2-1-1-7
☑1 ☐2 ☐3 ☑C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☑3 ☐C ☑H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H

Create	Discussion	Pair / group	Learner choice	Self-assessment	Peer assessment	SEN / low	Gifted
something new · 2-1-1-7	expressing ideas · 2-1-1-8	cooperative work · 2-1-1-8	independent work · 3-2-1-6	2-2-1-1	2-2-1-1	specific support · 2-1-1-3	specific stretch · 2-1-1-3
☐1 ☑2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☐C ☑H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

ALWAYS VISIBLE IN DELIVERY: Safe, respectful climate • every learner drawn in • purposeful use of time • smooth transitions • learner voice • specific feedback and next steps • students act on feedback

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Teacher	Subject	Grade / Section	Date / Period	Unit / Topic
Mr. Ashraf Elbialy	Computer Science (IGCSE 0478)	Grade 10 - Classes 10BR1 to 10BR3	Wk 7 (4-8 Oct 2026) - Period 2	Ch 8 - 8.2 Data types

Pacing (2-1-1-2): ☑ On plan ☐ Ahead ☐ Behind

Weeks ahead / behind: 0 **Week / lesson:** Week 7 / Lesson 2 of 2

Starting point (2-2-1-3): *What did my last check (quiz, exit ticket, homework) show — and what am I changing today because of it?*

Exit ticket: 80% defined constant; <- vs = confusion remains. Today: 3-min warm-up on assignment, then data types.

1 OUTCOMES — What must students learn?

Desired learning outcome(s) — *trace to the curriculum / standard and the table of specifications (ToS)*

Students identify and use INTEGER, REAL, CHAR, STRING and BOOLEAN, choose a suitable type for given data, and convert types in Python (int(), float(), str()). Syllabus 8.1.2; AO1, AO2.

Success criteria — *what will successful learning look / sound like?*

I can choose the correct data type for 8 data items and justify it; I can explain why input() needs converting before arithmetic; my program converts and calculates correctly.

Key knowledge / skill / understanding:

Data type decides what values are stored and which operations work; Python input() always returns a string.

Likely misconception / access or challenge need:

Phone numbers are integers; '5' + '5' = 10; REAL and INTEGER are interchangeable.

2 PROOF — What will students do to show they have learned it?

Three activities, each with a DIFFERENT strategy, resource and evidence tool. Then tick the Coverage Check on page 2.

ACTIVITY 1 *student task + evidence of learning*

Strategy: Cooperative learning **Time:** 10 min

What students will do:

Card sort in groups of 3: 16 data cards (age, price, grade letter, passport number, isRegistered...) placed under INTEGER, REAL, CHAR, STRING, BOOLEAN with a reason.

Evidence tool: Observation checklist

Assessment: ☑ Diagnostic ☐ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Card-sort sets; 5 type headers.

Support (low performers / SEN):

Mixed-language groups; examples on each header.

Stretch (fast learners / gifted):

Find 2 items that could fit 2 types and argue the best choice.

ACTIVITY 2 *student task + evidence of learning*

Strategy: Inquiry / investigation **Time:** 20 min

What students will do:

Investigate in Python: what do type('5'), '5'+'5', int('5')+5, float ('2.5'), int('2.5') do? Record results in a table, then fix a broken 'average mark' program.

Evidence tool: Portfolio / notebook work

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☑ Pair ☐ Group

Resources:

Python/Thonny; investigation sheet; broken average.py.

Support (low performers / SEN):

Investigation sheet with first 2 rows done.

Stretch (fast learners / gifted):

Explain the error from int('2.5') and write code that avoids it.

ACTIVITY 3 *student task + evidence of learning*

Strategy: Debate **Time:** 7 min

What students will do:

'Should a phone number be stored as an INTEGER?' Two teams argue, using leading zeros and +966 as evidence; class votes.

Evidence tool: Oral questioning

Assessment: ☐ Diagnostic ☑ Formative ☐ End

Grouping: ☐ Individual ☐ Pair ☑ Group

Resources:

Debate prompt; example numbers.

Support (low performers / SEN):

Sentence starters.

Stretch (fast learners / gifted):

Link the answer to validation from Chapter 7.

3 PREPARE — What do I need to do so students can succeed?

Set students up to succeed:

☑ Hook / prior learning ☑ Explain or model ☑ Success criteria shared

☑ Key vocabulary / examples ☑ Real-life connection ☑ Clear instructions / resources

☑ Praise, recognition and reinforcement planned

My setup / modelling: Time: 6 min

Warm-up: 3 lines with <- vs =. Hook: a form with name, age, height, grade letter and 'Paid?' - what kind of data is each? Model DECLARE with each type and Python type() in the shell. Show input() returning a string and int(input()).

What is new or innovative in this lesson? *(ETEC looks for this)*

Students investigate the computer's behaviour themselves and record evidence before the rule is given; a short debate turns a typical exam trap into reasoning.

Progress recorded / shared:

☑ Gradebook ☐ Parents ☑ Written feedback in notebook / portfolio

4 RESPOND — What will I do with what I see and hear?

If students struggle or finish early I will:

☑ Re-teach ☑ Regroup ☑ Give extra support

☑ Stretch ☑ Have students redo / improve

What I will watch for / likely responses:

Treating numbers with leading zeros as INTEGER; CHAR vs STRING confusion; TypeError from adding str and int; float precision surprise (0.1+0.2). Likely: 'Python has no CHAR' - explain how exams still use it.

Close / exit check *(summative + time for learners' questions)* Time: 7 min

Exit check: choose the data type for 5 new items and explain one; predict output of '3' * 2. Learners' questions.

Homework / platform *(e.g. Madrasati)*

Coursebook 8.2 activity questions; complete a data dictionary (identifier, type, example) for a school library system.

5 REFLECT — After the lesson

Complete within a day. This is reflective-practice evidence and feeds your next Starting point.

What worked — and how do I know?

Card sort sparked useful talk across language groups; Saudi, Egyptian and Indian students compared phone formats, which made the debate real. Investigation table done by 85%.

What will I change next time?

Give the investigation table on paper; switching windows slowed weaker students. Leave float precision as a stretch only.

Students needing follow-up:

10BR1: 3 students unsure about CHAR vs STRING; one SEN student needs the type-colour poster at the desk.

COVERAGE CHECK

Tick where each must-have happens (1 · 2 · 3 = activity, C = close, H = homework). An empty box is a gap to fix before teaching. Literacy and numeracy apply in every subject.

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☐1 ☐2 ☐3 ☐C ☑H	☑1 ☐2 ☑3 ☐C ☐H	☑1 ☑2 ☐3 ☐C ☐H	☐1 ☑2 ☐3 ☐C ☐H	☐1 ☐2 ☐3 ☑C ☐H	☑1 ☐2 ☐3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H	☑1 ☑2 ☑3 ☐C ☐H

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